Combating Noise Pollution: Policy, Exposure, and Welfare*

Takeru Sugasawa HRAFJ
Shinsaku Takikawa LSE
Takanao Tanaka UC Berkeley

October 2026

This paper provides the first evidence on how national noise-control policies affect noise exposure and welfare. We assemble two decades of largely untapped nationwide measurements of road-traffic noise from over 14,000 monitoring points in Japan and detailed data on noise-mitigation policies obtained through freedom-of-information requests. From 2002 to 2022, roadside noise fell by 2.8 decibels, equivalent to roughly a 20% decline in perceived loudness. We attribute about 70% of this decline to noise-control policies, particularly tighter vehicle noise-emission standards and the expansion of low-noise pavement. Residents are willing to pay 1.3% of residential land value for a one-decibel reduction in road-traffic noise. We estimate that Japan's national noise-control policies generated approximately 5 trillion yen (US$38 billion) in benefits over two decades. These benefits suggest substantial unrealized gains in countries with weaker noise-control policies, such as the United States.

Keywords: Noise pollution; environmental regulation; policy evaluation; nonmarket valuation

JEL Codes: Q53, Q58, R31, R41

*We thank Joseph Shapiro, James Sallee, Michael Anderson, and Sefi Roth, as well as seminar and conference participants at UC Berkeley, IPWSD (SIPA), LSE, Applied Regional Economics Workshop, Sophia University, and Society for Environmental Economics and Policy Studies for insightful and constructive comments. We acknowledge financial support from the Institute of Economic Research at Hitotsubashi University. All errors are our own. Emails: Sugasawa, sugasawa@hrf.or.jp; Takikawa, jamubin@gmail.com; Tanaka, takanao.tanaka.0323@gmail.com.

1 Introduction

Noise pollution—unwanted or harmful sound—is a pervasive environmental externality. Noise exposure causes annoyance and sleep disturbance and is linked to poorer health, impaired cognition, and lower productivity (WHO, 2018).[1] The problem is hardly new: ancient Roman writers complained about urban noise, and nineteenth-century thinkers such as Schopenhauer and Babbage wrote about its interference with thought and work (Juvenal, 1918; Schopenhauer, 2015; Babbage, 1864). Despite centuries of concern, harmful noise exposure remains widespread. Today, an estimated 47 million Americans and more than 110 million people in Europe are exposed to harmful transportation noise (Shkembi et al., 2026; EEA, 2025).

Governments around the world differ sharply in how they address noise pollution. The United States, for example, enacted the Noise Control Act in 1972 and the Quiet Communities Act in 1978 (U.S. Congress, 1972, 1978). Yet the federal government effectively withdrew funding for these laws in 1982 (EPA, 2025), and federal efforts to control noise have remained limited since then. Much of the practical responsibility for noise control now rests with state and local governments. This lack of policy coordination leaves many residents "without legal recourse to stop harmful noise exposures" (APHA, 2021).

Japan, by contrast, has actively combated noise pollution since the late 1990s. While Japan has regulated noise since the Noise Regulation Act of 1968, the 1995 Supreme Court ruling on Route 43 served as a catalyst for stronger action against road-traffic noise. After nineteen years of litigation, the Supreme Court held the Japanese government liable for traffic-noise damages for the first time in its history (Supreme Court of Japan, 1995). The government acknowledged that its efforts to reduce road-traffic noise had been insufficient and moved quickly to strengthen noise control nationwide (Interministerial Council, 1995). By the mid-2000s, the national government allocated roughly ¥40 billion (US$250 million) annually to noise-mitigation policies. Through extensive interviews with government officials, we identified key policy responses over the following two decades. The government established a nationwide system to monitor road-traffic noise, promoted the installation of noise-reducing infrastructure, and tightened vehicle noise-emission standards.

This paper provides the first evidence on how national noise-control policies affect noise exposure and welfare, focusing on road traffic in Japan.[2] National approaches to mitigating noise pollution differ starkly across countries. By studying Japan's experience, we offer lessons for policymakers seeking to reduce noise pollution elsewhere.

[1] See Guski et al. (2017); Basner and McGuire (2018) for evidence on annoyance and sleep disturbance, van Kempen et al. (2018); Greenhill (2026) for health effects, and Dean (2024) for cognition and productivity.

[2] In Europe, an estimated 92 million people are exposed to harmful levels of road-traffic noise, compared with 18 million from rail and 2.6 million from aircraft (EEA, 2025).

We assemble what is, to our knowledge, the most comprehensive longitudinal dataset of road-traffic noise measurements ever compiled, drawing on largely untapped administrative records from Japan's Continuous Monitoring of Road Traffic Noise (CMRTN) system. The system collects measurements in a "nationwide, continuous, and consistent" manner (MOE, 2022). Our harmonized dataset contains 46,544 measurements across 14,261 road segments from 2002 to 2022 and covers areas home to an estimated 37.3 million residents. We link these measurements to data on noise-mitigation infrastructure from the Road Environmental Census, which we obtained through a Freedom of Information Act request. The census records low-noise pavement, noise barriers, roadside buffer strips, and sound-absorbing panels on elevated structures from 2000 to 2010.

This paper answers four empirical questions. First, how did road-traffic noise change in Japan over the two decades from 2002 to 2022 and who benefited from the change? Second, to what extent did Japan's national noise-control policies—particularly tighter vehicle sound-emission standards and expanded noise-mitigation infrastructure—contribute to these changes? Third, what is residents' willingness to pay for quieter roads? Finally, what are the monetized benefits of Japan's noise-mitigation policies?

Our answers to all four questions are striking. First, we find that road-traffic noise declined substantially in Japan, although the rate of decline slowed over time. Between 2002 and 2022, noise levels fell by 2.8 decibels (dB), equivalent to about a 50% reduction in acoustic energy and a 20% reduction in perceived loudness. To put this decline in context, it is roughly comparable to halving traffic volume, lowering the speed limit from 50 to 30 km/h (FOEN, 2025), switching a robot vacuum from standard to quiet mode (Hales, 2024), or moving from station ticket gates to a coffee shop (Sueoka et al., 2009). Roughly two-thirds of this decline occurred between 2002 and 2012, the first half of our study period. Initially noisier locations experienced faster declines, indicating convergence in noise exposure across locations. However, we find no evidence of larger reductions in areas with a higher share of college graduates.

Second, Japan's noise-control policies account for about 70% of the observed decline in noise. Specifically, Japan tightened vehicle noise-emission standards from the late 1990s onward, and these changes account for about 25% of the decline. The rapid expansion of low-noise pavement accounts for 41%, and the installation of noise barriers for 7.5%. Because the government implemented many of these policies nationwide, geographic variation provides limited scope to identify their effects separately. We instead combine engineering estimates of each intervention's effect with observed changes in its prevalence. Low-noise pavement, for example, expanded from 28.0% to 68.1% of measurement sites between 2002 and 2022, while noise barriers expanded from 8.3% to 13.0%. For vehicle standards, we combine changes in permissible noise limits with the share of vehicles subject to the new standards and fleet

turnover. The model also accounts for changes in traffic volume, heavy-vehicle shares, and electric and hybrid vehicle adoption.

Third, residents living near roads are willing to pay 1.14% of residential land value for a 1 dB reduction in road-traffic noise. We estimate this willingness to pay using a hedonic design that exploits the replacement of conventional dense-graded asphalt with low-noise pavement, Japan's most widely adopted noise-mitigation intervention. Low-noise pavement reduces noise by about 2.8 dB and raises residential land prices within 100 m of the road by about 3.2%, implying that a 1 dB reduction in noise is capitalized into a 1.14% increase in land prices. Beyond 100 m, low-noise pavement has little detectable effect on residential land prices, likely because the associated noise reductions become less perceptible.

The benefits of low-noise pavement far exceed its incremental costs. We estimate a benefit–cost ratio of 10–14, roughly an order of magnitude higher than the ratios of 1.4–1.7 reported for noise barriers by Moretti and Wheeler (2025).[3] Roads require periodic repaving regardless, so the relevant cost is only the additional expense of using low-noise pavement instead of conventional dense-graded asphalt. Although low-noise pavement costs about 30% more, this extra expense is small relative to the resulting increase in nearby land values.

Finally, we estimate that Japan's noise-control policies between 2002 and 2022 generated approximately 5.0 trillion Japanese yen (US$38 billion) in monetized benefits. We obtain this estimate by combining the estimated willingness to pay per decibel with the policy-attributable reduction in noise and the value of affected residential land.

Our paper has an important policy implication: stronger noise-control policies may generate substantial unrealized gains in countries with weaker noise-control regimes, such as the United States. One potential avenue is tighter vehicle noise-emission standards. Japan's standards are harmonized with widely adopted United Nations standards, whereas U.S. federal standards cover medium- and heavy-duty trucks and motorcycles but not passenger cars (MOE, 2016; EPA, 2026). Stronger vehicle noise regulation could therefore help reduce road-traffic noise, although we do not observe the costs that automakers bear to comply with such standards. Low-noise pavement is another promising intervention and appears highly cost-effective in our setting.

This paper contributes to three strands of literature. First, we contribute to the literature on landmark national environmental laws by providing the first comprehensive evidence on the aggregate benefits of national noise-mitigation policies. Existing studies evaluate major regulations addressing air pollution (Chay and Greenstone, 2005; Deschênes et al., 2017; Shapiro and Walker, 2018), water pollution (Keiser and Shapiro, 2019; Keiser et al., 2026), hazardous waste (Greenstone and Gallagher, 2008), and land conservation (Frank et al.,

[3]Moretti and Wheeler report the marginal value of public funds, measured as property value appreciation divided by construction costs: 1.7 for constructed barriers and 1.4 for proposed barriers.

2025), and show that environmental regulation can substantially reduce environmental externalities and generate sizable social benefits.[4] We extend this literature to noise pollution, which the WHO has described as an "underestimated threat" and for which regulatory responses remain comparatively limited (WHO, 2011). Our paper examines Japan's intensified efforts to combat noise pollution since the late 1990s under the Noise Regulation Act.

We also contribute to the literature on noise-mitigation policies by, to our knowledge, being the first to document nationwide long-run changes in noise pollution and assess the contributions of different noise-mitigation interventions. Despite extensive evidence on the adverse effects of noise exposure, basic facts about noise pollution—how it has evolved over time and who has benefited from these changes—remain scarce. This gap partly reflects data limitations: the United States and Europe largely rely on modeled measures of noise exposure (BTS, 2022; EEA, 2025), providing limited direct evidence on changes in measured noise levels.[5] China conducts extensive nationwide noise monitoring, but site-level longitudinal data are not publicly available.[6] Using two decades of largely untapped nationwide data from Japan, we document long-run trends in noise pollution and how these changes vary across socioeconomic groups.[7] We also extend studies that evaluate individual noise-mitigation interventions, often through their effects on property values (Lindgren, 2021; Moretti and Wheeler, 2025), by combining our noise measurements with information on a broad set of policies to quantify their contributions to long-run reductions in road-traffic noise.

Finally, we contribute to the literature on households' willingness to pay for quieter environments by using directly measured roadside noise. Existing studies exploit variation in flight paths, airport operations, transportation infrastructure, and noise-mitigation interventions to estimate how noise is capitalized into property values (Boes and Nüesch, 2011; Winke, 2017; Ahlfeldt et al., 2019; von Graevenitz, 2018; Allroggen et al., 2025; Sugasawa et al., 2025; Lindgren, 2021; Moretti and Wheeler, 2025). These studies generally rely on modeled noise exposure to infer willingness to pay for a 1 dB reduction in noise.[8] We es-

[4]These benefits do not necessarily exceed regulatory costs. For example, Keiser and Shapiro (2019) find that Clean Water Act grants substantially improved water quality, but their estimated effects on housing values were smaller than the costs of the grants. They note that this comparison may omit important benefits not reflected in housing values.

[5]Comparisons across noise-mapping rounds require caution. The U.S. Bureau of Transportation Statistics notes that its 2014 prototype is not directly comparable with later releases and that changes in traffic-data reporting can generate apparent changes in noise (BTS, 2022). In Europe, methodological changes made the original 2017 and 2022 estimates non-comparable, so EEA trend assessments reconstruct the 2017 baseline using the 2022 methodology (Blanes et al., 2023, pp. 6, 11).

[6]Existing studies use measurements from only a few cities or aggregate monitoring data to the city level. For example, Han et al. (2018) use cross-sectional measurements from Shenzhen, while Xu et al. (2020) analyze city-level data for 111 cities over 1991–2017.

[7]Greenhill (2026) combines seismic data and machine learning to construct a detailed map of ambient noise exposure across the contiguous United States.

[8]Greenhill (2026) uses seismic measurements to construct local noise exposure and estimates its causal

timate the effects of low-noise pavement on both measured roadside noise and residential land values, allowing us to estimate willingness to pay per decibel of observed noise reduction. Because low-noise pavement is widely adopted, our estimates are directly relevant for evaluating noise control at scale.

effects on health.

References

Ahlfeldt, G. M., Nitsch, V., and Wendland, N. (2019). Ease vs. noise: Long-run changes in the value of transport (dis)amenities. *Journal of Environmental Economics and Management*, 98:102268.

Allroggen, F., Hansman, R. J., Knittel, C. R., Li, J., Wan, X., and Wang, J. (2025). Planes overhead: How airplane noise impacts home values. NBER Working Paper 34431, National Bureau of Economic Research.

APHA (2021). Noise as a public health hazard. Policy Statement 202115. Adopted October 25, 2021. Accessed September 24, 2026.

Babbage, C. (1864). *Passages from the Life of a Philosopher*. Longman, Green, Longman, Roberts, and Green, London. Chapter XXVI, Street Nuisances. A separately printed extract is available through the Wellcome Collection.

Basner, M. and McGuire, S. (2018). WHO environmental noise guidelines for the european region: A systematic review on environmental noise and effects on sleep. *International Journal of Environmental Research and Public Health*, 15(3):519.

Blanes, N., Fons, J., Hintzsche, M., Ramos, M. J., Sáinz, M., Milego, R., and Peris, E. (2023). Development of a 2017 baseline to monitor noise under the zero pollution objectives: Methodological document. ETC HE Report 2023/10, European Topic Centre on Human Health and the Environment. Published May 2024.

Boes, S. and Nüesch, S. (2011). Quasi-experimental evidence on the effect of aircraft noise on apartment rents. *Journal of Urban Economics*, 69(2):196–204.

BTS (2022). National transportation noise map documentation. Technical report, U.S. DOT. Prepared by the Volpe National Transportation Systems Center.

Chay, K. Y. and Greenstone, M. (2005). Does air quality matter? evidence from the housing market. *Journal of political Economy*, 113(2):376–424.

Dean, J. T. (2024). Noise, cognitive function, and worker productivity. *American Economic Journal: Applied Economics*, 16(4):322–360.

Deschênes, O., Greenstone, M., and Shapiro, J. S. (2017). Defensive investments and the demand for air quality: Evidence from the NOx budget program. *American Economic Review*, 107(10):2958–2989.

EEA (2025). Environmental noise in europe 2025. EEA Report 05/2025, EEA. Based on the 2022 Environmental Noise Directive reporting round. Online version corrected March 27, 2026.

EPA (2025). EPA history: Noise and the noise control act. Official agency history web page. Last updated November 6, 2025. Accessed September 24, 2026.

EPA (2026). Transportation equipment noise emission controls. 40 CFR Part 205, Subparts B and D. Electronic Code of Federal Regulations. Sections 205.50 and 205.150 specify applicability to medium and heavy trucks and motorcycles, respectively. Current version

accessed September 28, 2026.

FOEN (2025). Speed reductions. Federal Office for the Environment, Switzerland. https://www.bafu.admin.ch/en/speed-reductions. Published November 18, 2025.

Frank, E. G., Auffhammer, M., McLaughlin, D. W., Spiller, B., and Sunding, D. L. (2025). The cost of species protection: The land market impacts of the Endangered Species Act. Working Paper 25-01, Resources for the Future.

Greenhill, S. (2026). Noise pollution and infant health. Working paper, May 19.

Greenstone, M. and Gallagher, J. (2008). Does hazardous waste matter? evidence from the housing market and the Superfund program. *Quarterly Journal of Economics*, 123(3):951–1003.

Guski, R., Schreckenberg, D., and Schuemer, R. (2017). WHO environmental noise guidelines for the european region: A systematic review on environmental noise and annoyance. *International Journal of Environmental Research and Public Health*, 14(12):1539.

Hales, D. (2024). Lefant P1 review. Modern Castle. https://moderncastle.com/robot-vacuums/lefant-p1-review/. Updated February 19, 2024.

Han, X., Huang, X., Liang, H., Ma, S., and Gong, J. (2018). Analysis of the relationships between environmental noise and urban morphology. *Environmental Pollution*, 233:755–763.

Interministerial Council (1995). Policy for Implementing Countermeasures in Areas with Severe Road-Traffic Noise. https://www.env.go.jp/hourei/07/000045.html. Interministerial Liaison Council on Road Traffic Pollution Countermeasures. Joint directive of five ministries and agencies, December 1, 1995. In Japanese.

Juvenal (1918). *Satires*. William Heinemann, London. Translated by G. G. Ramsay, in Juvenal and Persius. Satire III, lines 232–238.

Keiser, D. A., Mazumder, B., Molitor, D., and Shapiro, J. S. (2026). Water works: Causes and consequences of safe drinking water in america. NBER Working Paper 35288, National Bureau of Economic Research.

Keiser, D. A. and Shapiro, J. S. (2019). Consequences of the clean water act and the demand for water quality. *Quarterly Journal of Economics*, 134(1):349–396.

Lindgren, S. (2021). A sound investment? traffic noise mitigation and property values. *Journal of Environmental Economics and Policy*, 10(4):428–445.

MOE (2002–2022). Continuous monitoring of road traffic noise results. Disseminated through the National Institute for Environmental Studies Environmental GIS. Available at https://tenbou.nies.go.jp/gisplus/download.html.

MOE (2016). Amendment to the permissible limits of motor vehicle noise. Ministry of the Environment, Japan, press release. March 18. In Japanese; title translated. Adopts the Phase 2 limits of UN Regulation No. 51, 03 series, for four-wheeled vehicles. Accessed September 28, 2026.

Moretti, E. and Wheeler, H. (2025). The traffic noise externality: Costs, incidence and policy implications. Technical report, National Bureau of Economic Research.

Schopenhauer, A. (2015). On noise and sounds. In *Parerga and Paralipomena: Short Philosophical Essays*, volume 2, chapter 30, pages 575–578. Cambridge University Press. Edited and translated by Adrian Del Caro. Originally published in 1851.

Shapiro, J. S. and Walker, R. (2018). Why is pollution from us manufacturing declining? the roles of environmental regulation, productivity, and trade. *American economic review*, 108(12):3814–3854.

Shkembi, A., Patel, K., Smith, L. M., Meier, H. C. S., and Neitzel, R. L. (2026). Racial and ethnic inequities to noise pollution from transportation- and work-related sources in the United States. *Journal of Exposure Science & Environmental Epidemiology*, 36:211–220. First published online July 17, 2025.

Sueoka, S., Uchida, H., Kikuchi, H., Kamoshida, H., Kadoya, M., and Tanaka, S. (2009). An investigation of the new index of the noise level in Japan. *Journal of Environmental Laboratories Association*, 34(4):254–261. In Japanese. https://tenbou.nies.go.jp/science/institute/region/journal/JELA_3404022_2009.pdf.

Sugasawa, T., Kuroda, Y., Nomura, K., Yasuda, S., and Yoshida, J. (2025). Urban responses to additional aircraft noise: Evidence from a natural experiment in Tokyo. Working paper, SSRN 5893696.

Supreme Court of Japan (1995). National Route 43 Pollution Litigation Judgment. Second Petty Bench, July 7, 1995; Minshu 49(7):1870 and 2599. In Japanese.

U.S. Congress (1972). Noise control act of 1972. Public Law 92-574, 86 Stat. 1234; codified as amended at 42 U.S.C. 4901 et seq.

U.S. Congress (1978). Quiet communities act of 1978. Public Law 95-609, 92 Stat. 3079.

van Kempen, E., Casas, M., Pershagen, G., and Foraster, M. (2018). WHO environmental noise guidelines for the european region: A systematic review on environmental noise and cardiovascular and metabolic effects: A summary. *International Journal of Environmental Research and Public Health*, 15(2):379.

von Graevenitz, K. (2018). The amenity cost of road noise. *Journal of Environmental Economics and Management*, 90:1–22.

WHO (2011). Noise. Fact sheet. December 1.

WHO (2018). *Environmental Noise Guidelines for the European Region*. WHO Regional Office for Europe, Copenhagen.

Winke, T. (2017). The impact of aircraft noise on apartment prices: A differences-in-differences hedonic approach for Frankfurt, Germany. *Journal of Economic Geography*, 17(6):1283–1300.

Xu, C., Zhang, Y., Cheng, B., Li, L., and Zhang, M. (2020). Study on environmental Kuznets curve for noise pollution: A case of 111 Chinese cities. *Sustainable Cities and Society*, 63:102493.